Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/22/2006
PI Name:	Marston, Philip L. Ph.D.		
Project Title:	Acoustic Radiation Pressure Control of Confined and Open Liquid Interfaces		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	01/08/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thiessen, David (Washington State University)		
Grant/Contract No.:	NNC04GA53G		
Performance Goal No.:			
Performance Goal Text:			
	This proposal is to investigate the use of acoustic radiation pressure to control and manipulate gas-liquid systems relevant to two-phase flow and interfacial stability in reduced gravity. In previous research [Marr-Lyon, Thiessen, & Marston, "Passive stabilization of capillary bridges in air with acoustic radiation pressure," Phys. Rev. Lett. 86, 2293-2296 (2001); Marr-Lyon, Thiessen, & P. L. Marston, "Stabilization of a cylindrical capillary bridge far beyond the Rayleigh-Plateau limit using acoustic radiation pressure and active feedback," J. Fluid Mech. 351, 345-357 (1997)] we have demonstrated that long liquid bridges may be stabilized by the application of suitable ultrasonic fields. This proposal will build upon that background to examine the ultrasonic response and/or control of other two-phase situations relevant to microgravity, including: (1) gas bubbles in bulk liquids; (2) stability of low-speed liquid jets and controlled breakup into drops; (3) liquid lenses (slugs), bubbles, and collars confined to tubes; and (4) liquid coatings (internal or external) on cylinders. Much of the research concerns the radiation pressure of ultrasonic guided waves and in some of		

Task Description:	the jet experiments, the wave is internal to the jet (with a frequency close to or above an acoustic waveguide cut-off frequency). Some related fundamental research may continue on new ways to analyze radiation pressure. The bubble and liquid jet experiments and some of the other experiments may also be appropriate for the KC-135. The primary objective is to see if the radiation pressure of ultrasound can be used to prevent difficulties that arise in two-phase gas-liquid contact in reduced gravity. The emphasis is on using radiation pressure of ultrasonic traveling waves to control gas or vapor bubbles in liquids. Other situations concern bubbles or particles in a liquid filled waveguides and the response to the radiation pressure of ultrasound guided by liquid jets. Other work concerns using the radiation pressure of sound to clear or move liquid slugs in a channel or confined passage. It is also plausible that acoustic radiation pressure and streaming may be used to enhance mixing and heat or mass transfer in capillary pipes. The radiation pressure of sound in a gas may also be used to control the thickness and stability of liquid films on a solid surface. The ability to control the dynamics of interfaces in the aforementioned situations will enhance the reliability of two-phase systems in reduced gravity. The current emphasis concerns situations where travelling ultrasonic waves may be used to control or remove bubbles. This may be important in critical applications. The extremely small value of the speed of sound compared to the speed of light means that the required ultrasound intensity is much smaller than the optical intensity needed to generate a given radiation stress. Experimental results suggest the method is likely to be useful in bulk liquids. Additional work is needed to determine efficient generation of radiation forces in waveguides.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While especially suitable for enhanced performance in reduced gravity two-phase systems, applications of this research in normal gravity are anticipated. These include, but are not limited to, medical ultrasonics (drug delivery and imaging), two-phase flow, and the control of liquid jets. It is also plausible that ultrasonic radiation pressure in waveguides may enhance the performance of certain laboratory diagnostic tests (details available upon request).
Task Progress:	There has been progress in FY 2006 in six areas. (1) We have developed and published a new way of calculating the ultrasonic radiation force on objects in ultrasonic beams and have applied it to the case of a Bessel beam. During the course of that research we discovered an unanticipated result that it is theoretically possible for the radiation force on an object to be directed opposite the direction of the ultrasonic beam such that the beam pulls on the object [P. L. Marston, J. Acoust. Soc. Am. 120, 3518-3524 (2006)]. (2) We have extended our prior calculations of radiation force based on acoustic scattering theory and have demonstrated equivalence with more restricted approaches [W. Wei and P. L. Marston, J. Acoust. Soc. Am. 118, 3397-3399 (2005)]. (3) We have significantly extended our prior work on the guiding of ultrasonic waves by liquid jets and the associated ultrasonic radiation force. (4) We have demonstrated some new aspects of transduction of ultrasonic guided waves. (5) We have completed and published work on the dynamics of capillary bridges started with the support of the expired grant NAG3-2378 [W. Wei, D. B. Thiessen, and P. L. Marston, Phys. Rev. E 72, 067304 (2005)]. (6) We have continued to investigate aspects of the ultrasonic radiation force on bubbles in liquids. Of these areas items, (1) and (6) may be significant to NASA since it may be possible to use radiation pressure to remove bubbles from liquids in critical low-gravity systems. It is feasible to use ultrasonic traveling waves in reduced gravity since only a small force is needed to move bubbles. The force can be much less than radiation forces used on the earth to levitate bubbles. Our experimental method enables the long-time responses of bubbles to ultrasonic traveling waves to be monitored without the experimental demands of reduced gravity. While radiation pressure of standing waves are often studied in normal gravity (since it is relatively easy with standing waves to generate large forces) ultrasonic traveling waves are more likely to be useful in low gravity. We developed a new type of dielectrophoretic bubble levitator which cancels the buoyancy of a bubble while simultaneously allowing exposure to traveling ultrasonic waves. We have improves our ability to generate ultrasonic beams in the dielectric liquid used in this system. Though our approach can also be used with standing waves, we have emphasized traveling waves since that is the most flexible approach to controlling or removing bubbles in reduced gravity. Research into Item (3) has given an improved understanding of drop pinch-off and break-up processes in low speed liquid jets as well as the ultrasonic modification of such processes. This work also has potential applications to stabilization and mixing and to clearing pipes and tubes.
Bibliography Type:	Description: (Last Updated: 06/21/2021)
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